

SUSPICIOUS ACTIVITY

Recommendation: RECOMMENDED: SECONDARY AUTHENTICATION & TRANSACTION REVIEW

RISK SCORE

34.42 / 100

MODEL DECISION

NORMAL

Transaction Overview & Account Baseline

Current Transaction	Evaluated Values	Customer Profile Baseline	Historical Norms
Transaction ID	TXN0000360089	Historical Total Txns	0
Customer ID	CUST00049761	Average Spend	\$114.27
Timestamp	2024-12-30 23:58:00	Spend Std Deviation	\$135.20
Amount	\$4.03 USD	Median Spend	\$75.76
Location	Mexico City, USA	Home City	Munich
Category	ATM Withdrawal	Home Country	USA
Payment Method	Mobile Payment	Primary Category	ATM Withdrawal
Device Type	POS Terminal	Primary Payment	Credit Card
Distance from Home	56.6 km	Primary Device	Desktop
Time Since Prior Txn	13.07 hrs	Avg Transaction Hour	13.8:00 hrs
Failed PIN Attempts	2	Txn Velocity	0.00 / day
Recent PIN Change	No	Customer Age	63 yrs
Cross-Border / Intl	No (Domestic)	Credit Score	721
Night Window	Yes (Off-hours)	Prior Confirmed Frauds	0

AI Model Risk Breakdown & Explainability

Autoencoder Anomaly Score		0.2801 (0.00 – 1.00)		Reconstruction MSE Error		0.332217 (Baseline Threshold: 0.5929)	
Risk Component	Score	Weight	Contrib (pts)	AI Explanation & Context			
Geographic Risk	0.80	15%	+12.00	Very long-distance transaction: 9847 km from Munich to Mexico City (13.1 hours since last transaction).			
Autoencoder	0.28	35%	+9.80	Transaction pattern appears relatively normal (autoencoder score: 0.28).			
Time Anomaly	0.70	10%	+7.00	Transaction at 23:00 — unusual off-hours activity (night transaction flag set).			
Velocity Risk	0.27	10%	+2.71	Transaction 13.1 h after last — within normal velocity range.			
Behavioral Deviation	0.19	8%	+1.50	Behavioural deviations detected: unusual city 'Mexico City', unusual payment method 'Mobile Payment', unusual device 'POS Terminal'. Customer normally uses Credit Card in Munich at avg \$114.			
Failed Attempts	0.67	2%	+1.33	2 prior failed attempts — moderate credential-stuffing risk.			
Credit Factor	0.06	1%	+0.06	Good credit score (721) — low credit risk.			
Amount Deviation	0.00	15%	+0.00	Transaction amount \$4.03 is 0.8σ below customer's average \$114.27 (σ=\$135.20).			
International Flag	0.00	3%	+0.00	Domestic transaction — no international risk.			
Pin Change	0.00	1%	+0.00	No recent PIN change.			

Audit & Compliance Sign-Off**Investigator Signature:**
_____**Verified Date:** 2026-10-07

This report is automatically synthesized by the SENTINEL Deep Autoencoder engine with 10-factor weighted anomaly scoring. All data points conform to regulatory audit guidelines.